

C# Tips & Tricks

SWEN1 | BIF3

Table of Contents

- Nullable Reference Types
- Pattern Matching
- Records
- Modern Syntax Features
- Visual Studio Shortcuts



Nullable Reference Types

Enabling null safety

```
string? email = null;
string name = "Joe";
```

- Helps catch possible `null` usage errors at compile time
- Enable in `.csproj`:

```
<Nullable>enable</Nullable>
```

- Use `?` for nullable, `!` to suppress warning, `??` or null checks for safety



Pattern Matching

`switch` Expressions

```
string message = userRole switch {
  "admin" => "Full access",
  "user" => "Limited access",
  _ => "No access"
};
```

- Replaces `switch` statement with concise expression
- More readable and expression-oriented code

Type & Property Patterns

```
if (person is Employee { Salary: > 5000 })
{
    Console.WriteLine("High earner");
}
```

- Checks both type and specific property values
- Allows destructuring and matching in one line



Records

Value-based equality

```
public record User(string Name, int Age);
```

- Immutable reference type with built-in `Equals()`, `ToString()`
- Compared by value, not by reference
- Ideal for DTOs, API models, configuration objects

With-Expressions

```
var user2 = user1 with { Name = "Anna" };
```

- Creates a new instance with modified fields
- Keeps original unchanged



Modern Syntax Features

File-scoped namespaces

```
namespace MyApp;

class Program { }
```

- Applies the namespace to the entire file
- No braces or indentation required
- Cleaner layout, default in new .NET templates

Top-level statements

```
Console.WriteLine("Hello World");
```

- Used in simple apps, scripts, demos
- Removes the need for `Main()` boilerplate

Global Usings

```
global using System.Text.Json;
```

- Declared once, used everywhere in the project
- Add to `GlobalUsings.cs`
- Reduces clutter in source files

Required Members

```
public class Config {
  public required string Url { get; init; }
}
```

- Forces required properties to be set during object creation
- Prevents runtime null errors
- Useful for configuration classes or API inputs



Visual Studio Shortcuts

Productivity Features

- `Ctrl + .` — Quick actions & refactorings
- `Ctrl + R, R` — Rename symbol
- `F12` — Go to definition
- `Ctrl + K, C` — Comment selection
- `Ctrl + K, U` — Uncomment selection
- `Ctrl + Shift + B` — Build project
- `F5` — Start with debugger
- `Ctrl + Alt + Q` — Open Quick Launch
- `Ctrl + ;` — Search files/types/symbols
- `Ctrl + Shift + Space` — Parameter info while typing

Further Reading

- C# Language Reference (Microsoft)
- What's New in C# 12
- Top-Level Statements
- Pattern Matching
- Required Members